AI-INTEGRATED SMART CAMPUS WITH WIRELESS SENSOR NETWORK

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1. Problem Statement

Classrooms, laboratories and other campus spaces use electricity for lighting, fans, projectors and other equipment, but monthly bills and occasional manual readings provide little visibility into **when, where or why** energy is being used. Lights or fans left running, deteriorating equipment and unexpected load spikes may go unnoticed. A low-cost room-level system is needed to measure loads continuously, learn normal consumption patterns and highlight unusual usage early.

2. Proposed Solution

We propose an AI-enabled classroom energy monitor combining a wireless sensor node, a backend service and a web dashboard. The prototype measures two separate low-voltage loads—a **red LED lighting circuit powered from a 5 V DC supply through a current-limiting resistor**, and a **5 V DC fan**—plus room temperature and humidity. Measurements are sent over Wi-Fi; an AI model forecasts short-term energy use and flags unusual differences between measured and predicted consumption.

3. Hardware Components

Component	Purpose
ESP32 DevKit V1 / MYOSA node	Reads sensors, adds timestamps and sends measurements over Wi-Fi. Firmware will be adapted to the MYOSA board's available GPIO and I²C interfaces for the demonstration.
2 × INA219 sensors	Measure voltage and current separately for the LED circuit and the fan; power is calculated from sensor readings.
DHT11 sensor	Measures ambient temperature and relative humidity to provide environmental context.
Red LED(s) + 1 kΩ current-limiting resistor per LED	Demonstrate the classroom lighting load. Each ordinary red LED is used with a series resistor on the 5 V DC supply; the LED itself is not assumed to be a 5 V component.
5 V DC fan	Demonstrates a small fan load measured by the second INA219.
2 × N-channel MOSFETs + 1N4007 diode	Switch the LED and fan loads; the diode is placed across the fan for flyback protection.
Regulated 5 V DC supply, breadboard and jumper wires	Power and interconnect the low-voltage loads. The prototype uses DC only and does not involve mains wiring.

4. System Architecture

INA219 ×2 + DHT11	ESP32 / MYOSA node	Wi-Fi JSON data	FastAPI backend	Database + AI model	Web dashboard

The data path connects physical measurements to useful decisions: sensors → controller → wireless transmission → server-side storage and AI → dashboard for staff.

5. Complete Hardware-to-Dashboard Workflow

Step 1 — Measure: INA219 #1 measures the LED circuit and INA219 #2 measures the 5 V DC fan circuit. The DHT11 records temperature and humidity. Each measurement is timestamped.

Step 2 — Collect and format: The ESP32 reads both INA219 modules over I²C and the DHT11 through a digital GPIO. It packages timestamp, voltage, current, calculated power, temperature and humidity into a structured JSON record.

Step 3 — Send over Wi-Fi: The ESP32/MYOSA node sends the JSON record using an HTTP request to a Python FastAPI backend. For the prototype demonstration, the backend can run on a laptop connected to the same Wi-Fi network.

Step 4 — Validate and store: FastAPI checks incoming fields and saves records in SQLite. The database stores sensor history, predictions and anomaly events; PostgreSQL can be considered when scaling to multiple rooms.

Step 5 — Forecast with AI: The backend uses recent historical readings and time-based features to estimate energy consumption for the next 10-minute interval with an XGBoost model. Environmental or load features can be used when available.

Step 6 — Detect unusual use: The system compares measured energy with the forecast. A sufficiently large difference is flagged as a possible anomaly and logged with timestamp and severity. The threshold will be calibrated using real classroom measurements.

Step 7 — Display on the dashboard: The browser dashboard requests current status, history, predictions and anomaly records from the backend. It presents recent readings, actual-versus-predicted charts, hourly/daily summaries and an anomaly log so staff can investigate possible wastage.

6. Software and AI Components

Embedded firmware: Arduino/C++ for the current ESP32 prototype, with pin/interface mapping adapted for the MYOSA board. **Backend:** Python FastAPI REST API. **Storage:** SQLite for development, with PostgreSQL as a future scalable option. **Dashboard:** browser interface with charts for readings, trends, forecasts and event history. **AI:** XGBoost for short-horizon energy forecasting and residual-based anomaly detection.

7. Expected Outcome and Validation

- **Visibility:** timestamped measurements for separate lighting and fan loads.
- **Early investigation:** forecast-aware alerts help staff inspect unexpected consumption or possible equipment issues.
- **Demonstration:** show a controlled change in the LED/fan load and verify that readings and anomaly events appear on the dashboard.
- **Next validation step:** collect real classroom data over time, recalibrate thresholds and evaluate forecasting performance before claiming field-level savings.

Expected final demonstration: a working low-voltage sensor node that measures the LED and fan loads, transmits data wirelessly, stores readings, produces an AI-based forecast, records possible anomalies and displays the results on a web dashboard.